

Table of Contents

1.0	Research Design	1
1.1	Research Problem	1
1.2	Research Objectives	1
1.3	Type of Research	2
1.4	Justification of Research Type	2
1.5	Data Collection Strategy.....	3
2.0	Data Cleaning and Preparation.....	4
2.1	Data Preprocessing	4
2.2	Handling Missing Values	5
2.3	Handling Outliers.....	6
2.4	Encoding Categorical Variables.....	6
2.5	Descriptive Statistics for Key Variables.....	6
3.0	Analytics Storytelling.....	8
3.1	Principles of Analytics Storytelling.	8
3.2	Narrative of Findings.....	8
3.3	Visualizations and Explanation.	9
4.0	Reporting and Presentation	12
4.1	Summary of Findings	12
4.2	Recommendations	12
5.0	Ethics and Legal Considerations.	14
5.1	Ethical Issues	14
5.2	Legal Considerations	14
5.3	Data Privacy, Compliance, and Fairness	15
6.0	Conclusion	16
7.0	References.....	17

1.0 Research Design

1.1 Research Problem

Academic achievement by students is one of the most significant indicators of educational achievement and may be conditioned by several other factors besides the ability. Gender, social economic background, and access to academic support are some aspects that can influence the performance of students in various subjects. These factors are also important to understand among the teachers and institutions that seek to enhance the learning outcomes. The dataset provides information on student performance on mathematics, reading, and writing, background measures, including gender, the type of lunch, and attendance in some course of test preparation. With these variables, it is possible to analyze the performance of different groups of students depending on different conditions. The problem of the research is whether these are related to differences in academic performance. In the absence of this analysis, it will be hard to decide what categories of students might need special assistance or focus. Thus, this research will be devoted to the analysis of tendencies in student academic performance and learn how background factors are connected with academic success.

1.2 Research Objectives

The key aim of the research is to examine performance of the students in three fundamental academic subjects and also determine the factors which are linked to the performance variability.

The objective of this research are as follows:-

- To test the performance of the students as a whole in mathematics, reading, and writing using descriptive statistics.
- To determine the association of lunch type with variation in student performance in subjects.
- The aim of the study is to determine the role of test preparation courses on academic performance.
- To make a comparison between the performance of male and female students in various subjects.

- To draw patterns and trends in the data that can be used to make data-driven suggestions on how to improve academic outcomes.

These are intended to steer the analysis and to make sure that the research does not go astray trying to find something of value in the data.

1.3 Type of Research

This research takes the **descriptive research design**. Descriptive research is applied to examine and summarize the data that is present and to determine the patterns, association and trends within existing data (GeeksforGeeks, 2026). It is not manipulative of variables or experimental. Rather, it dwells on what is being seen in the data set. In this case study, descriptive research is employed to investigate the effect of performance of students in various categories, which include gender, lunch type and the test preparation status. The intention is to establish a clear picture on how these factors are related to academic outcomes.

1.4 Justification of Research Type

This study should be conducted using descriptive research since the aim is to examine the existing data instead of determining the cause and effects of certain variables. The data employed in the present research is observational, that is, data are recorded in terms of student information without any experimental manipulations. Causality cannot be identified since no manipulation of variables is done. Rather, the research is aimed at finding patterns, connections among variables. Indicatively, the analysis may reveal higher scores among the students who had gone through test preparation but this does not necessarily mean that the test preparation has been the cause. The latter is an adequate method since it ensures that groups can be meaningfully compared, and the insight obtained can be used to make decisions. It is also compatible with the type of data, which is not experimental but is in a form of analysis.

1.5 Data Collection Strategy

The dataset that was used in this study is the secondary dataset, which is found to have the information on the student performances and the background variables associated with the student performances (GeeksforGeeks, 2026). It contains information about gender, race or ethnicity, parental level of education, type of lunch, a course in test preparation and mathematics, reading and writing scores. The dataset was chosen due to the availability of the relevant variables, which can be used to analyze both academic performance and potential factors that might have caused it. It is well structured, comprehensive, and descriptively analyzed. A sample of 500 observations was taken out of the original dataset to make the dataset manageable and in line with the requirements of the assignment. This is an efficient method of sampling, and at the same time gives adequate data to be used in analysis. The information was taken out of a trusted and public source and includes no personally identifiable information. This is suitable in terms of academic analysis besides reducing privacy issues.

2.0 Data Cleaning and Preparation.

2.1 Data Preprocessing

Preprocessing of data is a critical process of making sure that the dataset is correct, consistent and analysis worthy (Adeyeye, 2025). The data used in this research was initially entered into Microsoft Excel where it was saved in excel format to enable it to be manipulated and analyzed easily. To fit the assignment criteria, the dataset was narrowed down to 500 observations through sampling the first 500 lines of data. This measure will make the dataset manageable and still obtain adequate data to make meaningful analysis. Then, duplicate records were verified with the help of the Excel application in the “Remove duplicates” tool on all columns. The outcome indicated that no repeated records in the chosen data set were present, thus, attesting to the fact that the records are unique. The structure of the dataset was also checked so that all the variables were properly laid out. A column was a particular variable and each of these variables contained categorical variables, like the gender, the type of lunch, the type of course of test preparation, and the numerical variables, such as reading, writing and mathematics scores.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score											
2	male	group A	high school	standard	completed	67	67	63											
3	female	group D	some high school	free/reduced	none	40	59	55											
4	male	group E	some college	free/reduced	none	59	60	50											
5	male	group B	high school	standard	none	77	78	68											
6	male	group E	associate's degree	standard	completed	78	73	68											
7	female	group D	high school	standard	none	63	77	76											
8	female	group A	bachelor's degree	standard	none	62	59	63											
9	male	group E	some college	standard	completed	93	88	84											
10	male	group D	high school	standard	none	63	56	65											
11	male	group C	some college	free/reduced	none	47	42	45											
12	male	group E	some college	standard	completed	99	83	85											
13	female	group D	high school	standard	completed	80	87	90											
14	male	group D	associate's degree	standard	completed	77	87	85											
15	male	group C	high school	standard	completed	74	74	73											
16	male	group E	some high school	standard	completed	81	87	85											
17	male	group E	associate's degree	free/reduced	none	69	61	57											
18	male	group B	high school	standard	none	58	47	42											
19	female	group C	associate's degree	standard	completed	54	62	65											
20	female	group C	associate's degree	free/reduced	none	23	44	44											
21	male	group C	some college	free/reduced	none	39	32	31											
22	female	group C	associate's degree	standard	none	83	76	88											
23	male	group C	associate's degree	standard	none	58	52	55											
24	male	group B	some college	standard	completed	71	69	73											
25	female	group B	some college	free/reduced	none	44	55	54											

Figure 1: Dataset Imported into Microsoft Excel

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score											
479	male	group E	bachelor's degree	free/reduced	none	68	68	59											
480	male	group C	some college	free/reduced	completed	68	73	68											
481	male	group C	associate's degree	free/reduced	none	62	43	45											
482	female	group B	some high school	standard	none	76	88	84											
483	female	group C	bachelor's degree	free/reduced	none	77	79	83											
484	female	group E	master's degree	free/reduced	none	83	81	83											
485	female	group C	some college	free/reduced	none	71	78	82											
486	male	group D	some college	free/reduced	completed	86	91	89											
487	male	group B	master's degree	standard	none	93	96	96											
488	female	group B	associate's degree	free/reduced	none	60	64	60											
489	female	group D	some high school	standard	none	57	56	56											
490	male	group C	bachelor's degree	standard	none	84	79	74											
491	female	group D	bachelor's degree	standard	none	66	71	75											
492	male	group D	master's degree	standard	none	85	69	74											
493	female	group D	master's degree	standard	completed	57	68	76											
494	female	group C	some high school	standard	none	41	49	49											
495	female	group C	some high school	standard	none	75	76	76											
496	male	group D	high school	free/reduced	none	65	63	60											
497	male	group E	some college	standard	none	78	65	60											
498	female	group B	some high school	free/reduced	none	26	34	28											
499	male	group E	some college	free/reduced	none	71	64	66											
500	male	group B	bachelor's degree	standard	completed	86	92	93											
501	male	group B	associate's degree	standard	completed	89	84	81											
502																			

Figure 2: Dataset Reduced to 500 Rows

2.2 Handling Missing Values

Unobserved values may have adverse impacts on the validity of analysis and give misleading outcomes. Thus, missing data should be revealed and fixed in advance before analysis. The Excel filtering was used to check all the columns to give any blank or missing entries. The findings indicated that no values would be missing in the dataset. There was no need to do any imputation or data replacement since the data is complete. This would make sure that the analysis is not made based on assumptions or estimates and is made using complete and correct data.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score											
2	male	group A	high school	standard	completed														
3	female	group D	some high school	free/reduced	none														
4	male	group E	some college	free/reduced	none														
5	male	group B	high school	standard	none														
6	male	group E	associate's degree	standard	completed														
7	female	group D	high school	standard	none														
8	female	group A	bachelor's degree	standard	none														
9	male	group E	some college	standard	completed														
10	male	group D	high school	standard	none														
11	male	group C	some college	free/reduced	none														
12	male	group E	some college	standard	completed														
13	female	group D	high school	standard	completed														
14	male	group D	associate's degree	standard	completed														
15	male	group C	high school	standard	completed														
16	male	group E	some high school	standard	completed														
17	male	group E	associate's degree	free/reduced	none														
18	male	group B	high school	standard	none														
19	female	group C	associate's degree	standard	completed														
20	female	group C	associate's degree	free/reduced	none														
21	male	group C	some college	free/reduced	none														
22	female	group C	associate's degree	standard	none														
23	male	group C	associate's degree	standard	none														
24	male	group B	some college	standard	completed														
25	female	group B	some college	free/reduced	none														

Figure 3: Verification of No Missing Values in Dataset

2.3 Handling Outliers

Outliers can be defined as the values that are considerably greater or less than the bulk of the data. The lowest scores recorded in this dataset were not that high especially in mathematics. These values were not eliminated, however. All values were kept because it was known that low scores can be the actual performance of students and not data entry mistakes. These values were retained because they represent actual student performance rather than data errors. Removing them could distort the real distribution of the dataset.

2.4 Encoding Categorical Variables

The data set contains a number of categorical variables like gender, race or ethnicity, parental level of education, lunch type and test preparation course. Given that the analysis in this work is based on the descriptive statistics and visual comparison, the encoding of categorical variables into the numerical one was unnecessary. The pivot tables and visualizations directly used the categories to compare the performance of groups. The method will enable the analysis of the results clearly without the need to add unnecessary complexity to the process. This approach is appropriate because the analysis focuses on group comparisons rather than predictive modeling.

2.5 Descriptive Statistics for Key Variables.

The descriptive statistics give the summary of the performance of the students in three subjects: mathematics, reading and writing. The mean scores indicate that students had the highest scores in reading (68.76), then writing (67.29), and lastly mathematics (66.07). This shows that students usually scored higher in subjects that were language based whereas mathematics seemed to be more difficult. The lowest scores indicate differences in the performance of students. The lowest minimum score was in mathematics which stood at 13 as opposed to 28 in reading and 23 in writing. This implies that the poor performance was more evident in mathematics. The highest score among all the three subjects was 100 which indicated that the top students could achieve a high score in all subjects. The general findings show that there is fluctuation in student academic performance where mathematics exhibits a greater performance gap. This implies that there should be more academic facilitation in mathematics in order to enhance student performance.

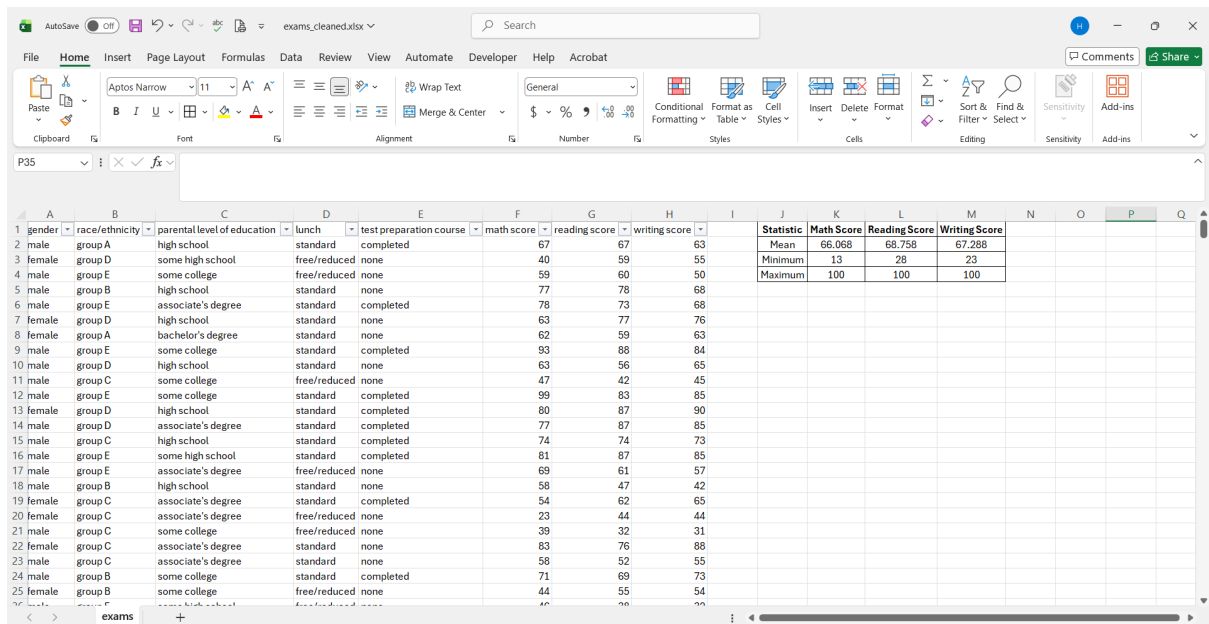


Figure 4: Summary of Descriptive Statistics for Key Variables

3.0 Analytics Storytelling

3.1 Principles of Analytics Storytelling.

The concept of analytics storytelling is concerned with the presentation of data in a manner that is easy to understand, meaningful, and clear to the non-technical stakeholders. It does not present data in a form of numbers, charts, or other forms but, rather, as a combination of data, visual representations, and narrative to explain insights and assist in decision-making. A good analytics narrative is logical. It begins by stating what the data reveals, why it is important and concludes with what needs to be done. This method aids the readers to grasp the relevance of the findings without the need of technical expertise. Clarity is another important principle. The visualizations must be straightforward, pertinent, and must be directly connected to the research objectives. Every chart must convey a particular message and the description must help the reader to gain the depth of what the data was revealing. It is also crucial that it remains consistent. The results provided on charts should correspond with the description. This makes the analysis dependable and simple to read.

3.2 Narrative of Findings

The dataset analysis indicates that there are some key trends in student performance in their academic studies. These tendencies are founded on comparisons among various groups of students such as the status of test preparation, type of lunch, and gender. A major discovery is associated with preparation of tests. Students that were provided with the preparation test always got higher average scores approximately 10–15 points in all subjects. They had a significantly better performance in mathematics, reading, and writing than those students who did not take the course. This is an indication that academic preparation is significant in enhancing academic performance. The other important observation is the disparity between the performances on the basis of lunch type. The students who had standard lunch had high average scores in all the three courses than those with either free or reduced lunch. Such a gap means that the academic performance can be affected by the external variables like access to resources and living conditions. Less privileged students might also encounter more complications that will influence their performance in exams. There are also performance differences between the male and female students as indicated in the analysis. Male students excelled in mathematics whereas the female students achieved higher marks in reading and writing. This trend implies that students might not be strong in every subject, and a special approach to studying could

help them to overcome the gaps. In general, the results indicate that academic performance is determined by various factors such as academic support as well as background conditions. Such insights will give a better picture of the performance of various students groups and the ways of how this can be improved.

3.3 Visualizations and Explanation.

Three Power BI visualizations were drawn to aid the analysis. These charts are a clear and structured way of presenting the data and hence, the comparisons across the various groups are made easier. The initial visualization represents average scores depending on the type of lunch. It also points out the disparity between performance of students with normal lunch and those with free or reduced lunch. This chart is a clear indication that students who have normal lunch record some of the highest scores in all subjects. The second visualization gives an average score by gender. It reveals that the male students have an average score that is higher in mathematics whereas the female students are better at reading and writing. This image allows one to readily compare subject-specific performance based on gender. The third visualization describes average scores according to the state of test preparation. It is evident that students that took the test preparation course perform better than students that did not take the course in all the three subjects. This supports the claim that academic preparation can enhance the performance. These visualizations are used to support the story, as they offer solid evidence of the trends that were identified during the analysis. They enable the non-technical stakeholders to see the most important findings fast and make knowledgeable choices about the information. These visualizations were created using clustered column and bar charts to enable clear comparison across categories. These findings highlight key areas where targeted academic interventions can improve student performance and reduce achievement gaps.

Average Scores by Lunch Type

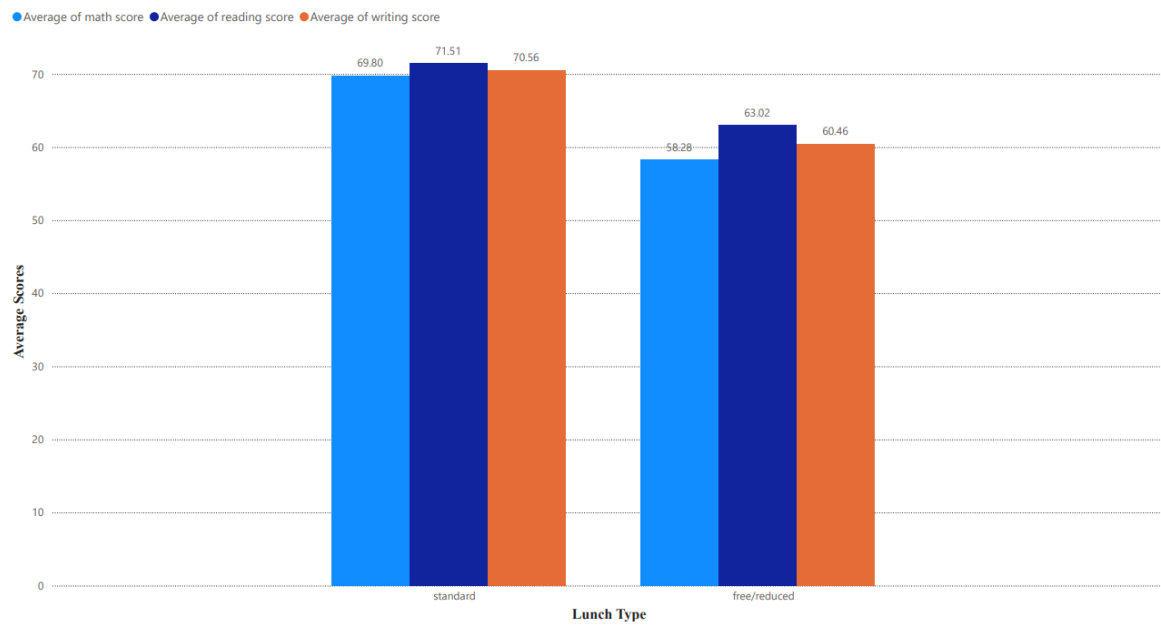


Figure 5: Power BI Visualization of Scores by Lunch Type

Average Scores by Gender

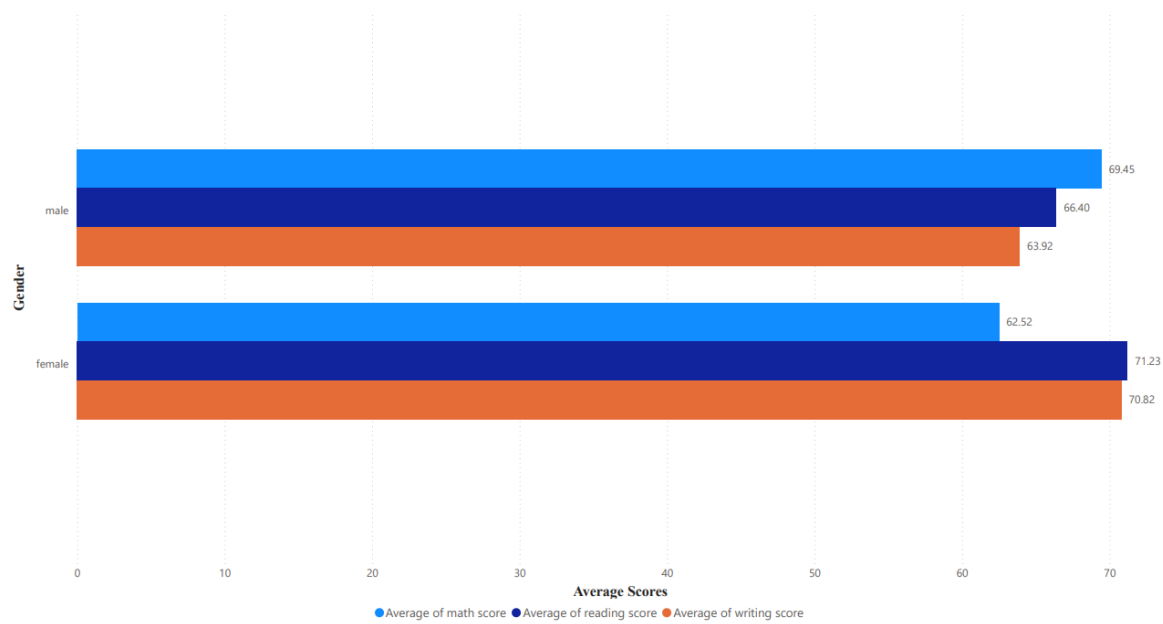


Figure 6: Power BI Visualization of Scores by Gender

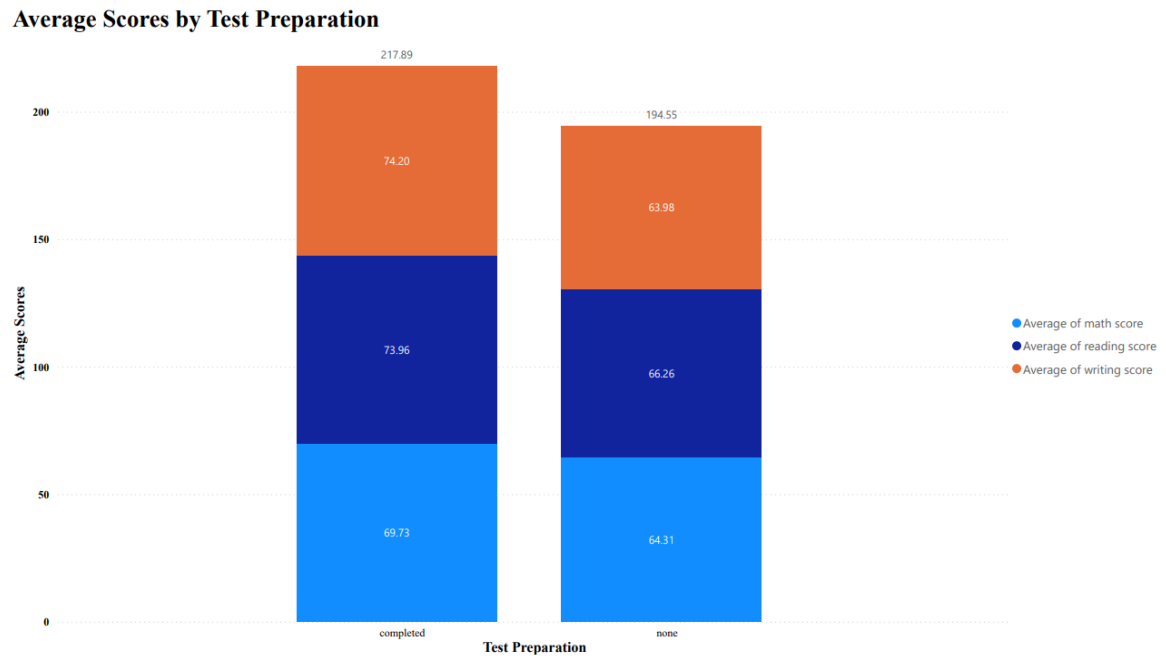


Figure 7: Power BI Visualization of Scores by Test Preparation

4.0 Reporting and Presentation

4.1 Summary of Findings

The analysis provides a clear understanding of student performance and the factors influencing it. The findings indicate that the performance of students depends on subjects and is dependent on a number of factors. The highest score in students was achieved in reading, and the lowest score was in mathematics. This shows that mathematics is mostly more difficult to the students than language based ones. The analysis indicates also that students who attended the test preparation course were better than the others in all subjects. This underlines the role of organized academic support to enhance performance of students. Moreover, a significant difference was noted depending on the type of lunch. The students who had a standard lunch always performed highly as compared to those who had free or reduced lunch. It implies that socioeconomic factors can also be a contributor of academic achievement. Differences were also found to be gender-based. Male students excelled more in mathematics compared to the female counterparts who excelled in reading and writing. This implies that students can exhibit various strengths in subjects. In general, the results indicate that academic achievement is not specifically a result of individual aptitude, but there is also the association with the resource availability, preparation, and background.

4.2 Recommendations

Resting on the results, a number of recommendations can be introduced that will help to enhance student academic performance. To begin with, educational institutions are supposed to promote the involvement in test preparation programs. The comparison reveals that students who had been enrolled in such programs always scored higher. Schools should implement organized revision, practice tests and directed learning to enable the students to prepare better. Second, students with lower socioeconomic background should be given additional support. The findings indicate that students who have free or reduced lunch have a tendency of performing poorly as compared to those who have regular lunch. Some of the ways through which schools can solve this is by providing academic support programs, mentoring, and availability of learning materials. Third, specific support must be carried out according to the subject performance. As mathematics is less averaged and more varied, more attention should be paid to enhancing mathematics skills. This may involve additional classes, tutoring, and interactive learning styles. Fourth, instructional methods must take into account various

learning abilities in the student. The female students had higher results in reading and writing compared to the male students which were higher in mathematics. Teachers are able to develop balanced instructional strategies that serve every learner and enable them to gain in areas of weaknesses. Lastly, decision-making in learning institutions should be driven by data. Data on performance of the students enables the schools to analyze the trends and track the progress and take the interventions in time to achieve better results.

5.0 Ethics and Legal Considerations.

5.1 Ethical Issues

To make sure that the analysis is responsible, the use of data in research has to be cautiously considered concerning the ethical principles. Even though these attributes do not involve a personally identifiable information, the dataset that was used in the given study still contains such sensitive attributes as gender, race or ethnicity and such socioeconomic indicators as lunch type. The danger of misinterpretation or bias is one of the ethical concerns. Unless the findings are elaborated with caution, they can cause undue generalization about some categories of students. As an illustration, when there are differences in performance, depending on gender or socioeconomic status, it should not be considered as a set characteristic, but as a tendency that can be manipulated by other forces. Fairness in analysis is also another issue. The research should make sure that no groups are not accurately represented and that the conclusions should not be made on assumptions, but on facts. This analysis must be geared towards enhancement and awareness, and not to mention or discriminate against any group. Moreover, ethical research involves transparency. Every action done in data cleaning, analysis, and interpretation should be well written in a way that the results can be interpreted and analyzed accordingly.

5.2 Legal Considerations

In legal terms, data manipulation should be in accordance with the principles of data protection. Work regulation in Malaysia is stipulated in the Personal Data Protection Act (PDPA), which suggests individual responsibility in the management of personal data. Even though this data is publicly accessible and anonymized, it is necessary to adhere to general data protection best practices. No personal information like names, identification numbers, or address is included in the dataset that was used in the study. This minimizes the possibility of privacy breaches and renders the dataset appropriate in the academic analysis. It is also important to cite the source of data properly. Citation of data without crediting its origin may cause problems of intellectual property and academic dishonesty. Thus, the database has to be suitably cited in the report.

5.3 Data Privacy, Compliance, and Fairness

The analysis is performed with the help of aggregated data instead of the identification of a person to guarantee the privacy of data. The implication of this is that the results are represented in form of groups, say, average scores, and are not displayed individually, by individual students. The adherence is achieved through the application of a publicly accessible dataset and through the assurance that one should not be trying to re-identify people. Every data is utilized solely on academic grounds. Fairness will be considered in respect of putting the results into perspective and not making biased conclusions. The findings are utilized to point at the points where support might be necessary, but not to judge or categorize the students as bad. In general, the ethical and legal issues are incorporated in the course of the study to make the analysis responsible, respectful, and in line with good research practices.

6.0 Conclusion

This research examined mathematics, reading and writing performance of students through an organized research method. The results indicate evident variability between subjects and among groups of students. The reading and writing scores were both highest and mathematics had lower average scores and a greater variation. This puts into perspective the necessity to put more attention on enhancing mathematics performance. It is also analyzed that students who attended test preparation scored higher in all subjects. This aids the need of organized educational assistance in enhancing performance. There were also differences depending on the lunch type. The students who had the regular lunch performed higher than the students with free lunch or reduced lunch and this means that external factors like access of resources could affect the performance of the students. Gender differences were also determined. Male students had higher results in mathematics whereas female students scored higher in reading and writing. This implies that students possess various areas of strengths in diverse subjects. The general findings of the study are that academic performance depends on various factors among them being preparation, background conditions, and learning strengths. These results denote the role of special assistance, an equal right to the resources, and data-driven decision-making in enhancing student outcomes.

7.0 References

Adeyeye, S. A. (2025). Introduction to research methods: A concise and practical approach for students and researchers, Megatrend Publications, Ibadan, Nigeria.

Alex The Analyst. (2020). Data analysis full course for beginners
https://www.youtube.com/watch?v=I0vQ_VLZTWg

Florence, C. (2024). Data storytelling in marketing: How to tell persuasive stories through data, 1st edition, Kogan Page.

GeeksforGeeks. (2026). Introduction to Power BI dashboard.
<https://www.geeksforgeeks.org/power-bi/introduction-to-power-bi-dashboard/>

GeeksforGeeks. (2026). Methods of data collection.
<https://www.geeksforgeeks.org/data-analysis/methods-of-data-collection/>

Ken Jee. (2020). Data analyst roadmap (what you need to know)
https://www.youtube.com/watch?v=jxq4-KSB_OA